

AR 518: Advanced Blockchain and web3

Course Code	AR 518
Course Name	Advanced Blockchain and web3
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Undergraduate (Elective) / Postgraduate
Approval	56th BoA (Jul 2024)

Course Contents

- **Advanced Ethereum concepts** (8 Lectures)
 - EIPs (Ethereum Improvement protocols)
 - Top EIPs that have changed the course of Ethereum
 - ERC standards (Ethereum Request for Comment)
 - Top Token standards
 - Smart contract security
 - Hardhat & Truffle
- **Layer 2's, Rollups, DeFi, Web3** (12 Lectures)
 - TLayer 2 chains: What is a Layer 2 chain
 - Differences between L1 and L2 chains
 - Techniques for Layer2 sidechain, State channels, Plasma, Rollups
 - Popular Layer2 chains – Polygon: setting up the wallet and deploying contracts to Layer 2 chains
 - Rollups : What is a Rollup, Optimistic Rollups, ZK Rollups
 - DeFi: Tokenization, DEX, Liquidity markets (AMM & bootstrapping), Stablecoins, Creating and deploying tokens and other smart contracts
 - Web3
 - DAO's
 - Metaverse
- **Third generation Blockchains** (8 Lectures)
 - Third-Generation Blockchain and Major Examples
 - Solana: Introduction, Architecture, consensus, Transaction policy
 - Solana Nodes and Network, Sample transactions
 - Cosmos: Introduction, IBC, Consensus and Network
 - Cosmwasm
- **DEPIN Decentralized Physical Infrastructure Networks** (2 Lectures)
 - Introduction to DEPIN networks
 - Filecoin, Storj, Arweave
 - Render Network
 - Theta Network
 - Helium
- **Hyperledger Fabric** (4 Lectures)
 - Hyperledger Fabric Architecture
 - Setting up and managing a Hyperledger network
 - Chain code development and deployment
- **Practicals, Projects and Assignments** (8 Lectures)
 - Create an ERC20 token, deploy and interact with it
 - Create a contract for Asset tokenisation or real-world scenario like crowdfunding/voting
 - Create Supply Chain Management / Healthcare Records management / Asset Tokenization on Hyperledger fabric

Text Books

1. **Mastering Ethereum: Building Smart Contracts and DApps**
2. **Blockchain with Hyperledger Fabric**, 2nd Edition

References

1. **Practical Limitations of Ethereum's Layer-2** RAY NEIHEISER^{1,2}, GUSTAVO INACIO³, LUCIANA RECH³, CARLOS MONTEZ², MIGUEL MATOS¹ and LUÍS RODRIGUES¹
2. Xiangyu Li, Xinyu Wang, Tingli Kong, Junhao Zheng, and Min Luo, **From Bitcoin to Solana – Innovating Blockchain Towards Enterprise Applications**.
3. Kim S, Deka GC, editors, **Advanced applications of blockchain technology**, Springer, 2020.